

Air University



# Network Security (Lab)

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Mr. Haseeb Ahmed

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Department of Cyber Security

USAMA ARSHAD | 247141

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## Lab Task # 01

### 1. Checking IP configuration

**Command:** ipconfig

**Explanation:** identifies unauthorized IP or gateway assignments. Ensure the device is connected to the intended, secure network assignments.

```
Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::535b:5eee:ed52:f2d0%19
    IPv4 Address. . . . . : 172.18.18.204
    Subnet Mask . . . . . : 255.255.252.0
    Default Gateway . . . . . : 172.18.16.1

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
```

### 2. View Detailed Information

**Command:** ipconfig /all

**Explanation:** Exposes MAC addresses and DNS settings to detect hardware spoofing. Vital for identifying DNS hijacking where traffic is sent to malicious servers.

```
Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 
    Description . . . . . : Bluetooth Device (Personal Area Network)
    Physical Address. . . . . : 24-41-8C-1F-81-EF
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    Description . . . . . : Intel(R) Wi-Fi 6 AX200 160MHz
    Physical Address. . . . . : 24-41-8C-1F-81-EB
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::535b:5eee:ed52:f2d0%19(Preferred)
    IPv4 Address. . . . . : 172.18.18.204(Preferred)
    Subnet Mask . . . . . : 255.255.252.0
    Lease Obtained. . . . . : Wednesday, February 11, 2026 1:04:56 PM
    Lease Expires . . . . . : Wednesday, February 11, 2026 9:04:55 PM
    Default Gateway . . . . . : 172.18.16.1
    DHCP Server . . . . . : 172.18.16.1
    DHCPv6 IAID . . . . . : 220479884
    DHCPv6 Client DUID. . . . . : 00-01-00-01-2E-94-98-02-38-22-E2-3E-6D-AD
    DNS Servers . . . . . : 172.18.0.201
                           8.8.8.8
    NetBIOS over Tcpip. . . . . : Enabled
```

### 3. Test Network Connectivity

**Command:** ping google.com

**Explanation:** Diagnose host reachability to identify potential network outages. Significant delays or packet loss can indicate a targeted Denial of Service (DoS) attack.

```
C:\Users\HP>ping www.google.com

Pinging www.google.com [142.250.201.228] with 32 bytes of data:
Reply from 142.250.201.228: bytes=32 time=50ms TTL=116
Request timed out.
Reply from 142.250.201.228: bytes=32 time=167ms TTL=116
Reply from 142.250.201.228: bytes=32 time=64ms TTL=116

Ping statistics for 142.250.201.228:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 50ms, Maximum = 167ms, Average = 93ms
```

### 4. Trace Network Path

**Command:** tracert google.com

**Explanation:** Maps the path to a destination to reveal suspicious intermediate hops. Helps detect “man in the middle” interceptions or unauthorized traffic redirection.

```
C:\Users\HP>tracert google.com

Tracing route to google.com [142.251.37.142]
over a maximum of 30 hops:

  1    15 ms    20 ms    6 ms    172.18.16.1
  2     9 ms     4 ms     2 ms    10.100.100.1
  3    11 ms     3 ms     3 ms    103.4.95.217.pern.pk [103.4.95.217]
  4    14 ms    225 ms    11 ms    172.31.65.141
  5   276 ms    451 ms    59 ms    117.20.21.61
  6    31 ms     26 ms    31 ms    110.93.254.33
  7    61 ms     35 ms    32 ms    110.93.252.48
  8    50 ms    136 ms    53 ms    110.93.252.136
  9    52 ms     48 ms    52 ms    110.93.192.207
 10   152 ms    106 ms    44 ms    172.253.51.55
 11     *        40 ms    42 ms    142.251.51.57
 12    66 ms     64 ms   124 ms    pnfjra-aj-in-f14.1e100.net [142.251.37.142]

Trace complete.
```

## 5. View Active Network Connections

**Command:** netstat

**Explanation:** Lists active connections to find unauthorized open ports. Useful for spotting “phone home” activity from installed malware or backdoors.

```
C:\Users\HP>netstat

Active Connections

Proto Local Address           Foreign Address         State
TCP    127.0.0.1:49670          ELITEBOOK:49671        ESTABLISHED
TCP    127.0.0.1:49671          ELITEBOOK:49670        ESTABLISHED
TCP    127.0.0.1:53244          ELITEBOOK:53246        ESTABLISHED
TCP    127.0.0.1:53246          ELITEBOOK:53244        ESTABLISHED
TCP    172.18.18.204:53541      4.213.25.241:https      ESTABLISHED
TCP    172.18.18.204:53698      4.213.25.240:https      ESTABLISHED
TCP    172.18.18.204:53978      52.123.243.183:https    ESTABLISHED
TCP    172.18.18.204:53999      40.99.26.178:https      ESTABLISHED
TCP    172.18.18.204:54000      40.99.26.178:https      ESTABLISHED
TCP    172.18.18.204:54001      a23-210-96-161:http     CLOSE_WAIT
TCP    172.18.18.204:54003      13.107.246.254:https    CLOSE_WAIT
TCP    172.18.18.204:54130      dns:https               ESTABLISHED
TCP    172.18.18.204:54182      13.107.139.11:https     TIME_WAIT
^C
```

**Command:** netstat -an

```
TCP    172.18.18.204:54231      52.182.143.215:443      ESTABLISHED
TCP    172.18.18.204:54232      13.107.139.11:443       ESTABLISHED
TCP    192.168.74.1:139         0.0.0.0:0               LISTENING
TCP    192.168.180.1:139        0.0.0.0:0               LISTENING
TCP    [::]:135                 [::]:0                  LISTENING
TCP    [::]:445                 [::]:0                  LISTENING
TCP    [::]:623                 [::]:0                  LISTENING
TCP    [::]:5357                [::]:0                  LISTENING
TCP    [::]:16992               [::]:0                  LISTENING
TCP    [::]:49664               [::]:0                  LISTENING
TCP    [::]:49665               [::]:0                  LISTENING
TCP    [::]:49666               [::]:0                  LISTENING
TCP    [::]:49667               [::]:0                  LISTENING
TCP    [::]:49668               [::]:0                  LISTENING
TCP    [::]:49689               [::]:0                  LISTENING
TCP    [::1]:49669              [::]:0                  LISTENING
UDP    0.0.0.0:500              *:.*                     *.*
UDP    0.0.0.0:3702             *:.*                     *.*
UDP    0.0.0.0:3702             *:.*                     *.*
UDP    0.0.0.0:4500             *:.*                     *.*
UDP    0.0.0.0:5050             *:.*                     *.*
UDP    0.0.0.0:5353             *:.*                     *.*
UDP    0.0.0.0:5355             *:.*                     *.*
UDP    0.0.0.0:58651            *:.*                     *.*
UDP    0.0.0.0:63712            *:.*                     *.*
UDP    127.0.0.1:1900           *:.*                     *.*
UDP    127.0.0.1:50800          *:.*                     *.*
UDP    127.0.0.1:52352          *:.*                     *.*
```

## 6. Identify processes using Network Ports

**Command:** netstat -ano

**Explanation:** Lists network traffic to specific process IDs (PIDs) for deep investigation. Allows admins to identify and terminate the exact malicious process causing a connection.

|     |                   |           |           |       |
|-----|-------------------|-----------|-----------|-------|
| TCP | 192.168.74.1:139  | 0.0.0.0:0 | LISTENING | 4     |
| TCP | 192.168.180.1:139 | 0.0.0.0:0 | LISTENING | 4     |
| TCP | :::135            | :::0      | LISTENING | 1128  |
| TCP | :::445            | :::0      | LISTENING | 4     |
| TCP | :::623            | :::0      | LISTENING | 5780  |
| TCP | :::5357           | :::0      | LISTENING | 4     |
| TCP | :::16992          | :::0      | LISTENING | 5780  |
| TCP | :::49664          | :::0      | LISTENING | 972   |
| TCP | :::49665          | :::0      | LISTENING | 820   |
| TCP | :::49666          | :::0      | LISTENING | 1620  |
| TCP | :::49667          | :::0      | LISTENING | 2160  |
| TCP | :::49668          | :::0      | LISTENING | 4452  |
| TCP | :::49689          | :::0      | LISTENING | 940   |
| TCP | :::1:49669        | :::0      | LISTENING | 5840  |
| UDP | 0.0.0.0:500       | *:*       |           | 5372  |
| UDP | 0.0.0.0:3702      | *:*       |           | 15376 |
| UDP | 0.0.0.0:3702      | *:*       |           | 15376 |
| UDP | 0.0.0.0:4500      | *:*       |           | 5372  |
| UDP | 0.0.0.0:5050      | *:*       |           | 7208  |
| UDP | 0.0.0.0:5353      | *:*       |           | 3376  |
| UDP | 0.0.0.0:5355      | *:*       |           | 3376  |
| UDP | 0.0.0.0:58651     | *:*       |           | 15376 |
| UDP | 0.0.0.0:63712     | *:*       |           | 6840  |
| UDP | 127.0.0.1:1900    | *:*       |           | 7072  |
| UDP | 127.0.0.1:50800   | *:*       |           | 4800  |

## 7. Display ARP Table

**Command:** arp -a

**Explanation:** Checks the IP-to-MAC mapping table for inconsistencies. It is the primary way to detect arp poisoning where an attacker intercepts local data.

```
C:\Users\HP>arp -a

Interface: 192.168.74.1 --- 0x7
    Internet Address      Physical Address        Type
    192.168.74.255        ff-ff-ff-ff-ff-ff      static
    224.0.0.22            01-00-5e-00-00-16      static
    224.0.0.251           01-00-5e-00-00-fb      static
    224.0.0.252           01-00-5e-00-00-fc      static
    239.255.255.250       01-00-5e-7f-ff-fa      static

Interface: 192.168.180.1 --- 0xc
    Internet Address      Physical Address        Type
    192.168.180.255       ff-ff-ff-ff-ff-ff      static
    224.0.0.22            01-00-5e-00-00-16      static
    224.0.0.251           01-00-5e-00-00-fb      static
    224.0.0.252           01-00-5e-00-00-fc      static
    239.255.255.250       01-00-5e-7f-ff-fa      static

Interface: 172.18.18.204 --- 0x13
    Internet Address      Physical Address        Type
    172.18.16.1           54-16-51-60-08-d6      dynamic
    172.18.19.255         ff-ff-ff-ff-ff-ff      static
    224.0.0.2            01-00-5e-00-00-02      static
    224.0.0.22            01-00-5e-00-00-16      static
    224.0.0.251           01-00-5e-00-00-fb      static
    224.0.0.252           01-00-5e-00-00-fc      static
    239.255.255.250       01-00-5e-7f-ff-fa      static
    255.255.255.255       ff-ff-ff-ff-ff-ff      static
```

## 8. Check DNS Resolution

**Command:** nslookup google.com

**Explanation:** Verifies domain-to-IP resolution to ensure records haven't been tampered with. Essential for identifying DNS cache poisoning or phishing redirects.

```
C:\Users\HP>nslookup google.com
Server: AD-Faculty.faculty.local
Address: 172.18.0.201

Non-authoritative answer:
Name: google.com
Addresses: 2a00:1450:4019:812::200e
          142.250.202.14
```

## 9. View Routing Table

**Command:** route print

**Explanation:** Examines the internal routing table for unauthorized path changes. Detects hidden routes that might be bypassing security firewalls or filters.

```
C:\Users\HP>route print
=====
Interface List
18...38 22 e2 3e 6d ad .....Intel(R) Ethernet Connection (6) I219-LM
15...24 41 8c 1f 81 ec .....Microsoft Wi-Fi Direct Virtual Adapter
 3...26 41 8c 1f 81 eb .....Microsoft Wi-Fi Direct Virtual Adapter #2
 7...00 50 56 c0 00 01 .....VMware Virtual Ethernet Adapter for VMnet1
12...00 50 56 c0 00 08 .....VMware Virtual Ethernet Adapter for VMnet8
 6...00 ff 43 b8 30 38 .....TAP-Windows Adapter V9
19...24 41 8c 1f 81 eb .....Intel(R) Wi-Fi 6 AX200 160MHz
14...24 41 8c 1f 81 ef .....Bluetooth Device (Personal Area Network)
 1.....Software Loopback Interface 1
=====
```

## 10. Test localhost Connectivity

**Command:** ping 127.0.0.1

**Explanation:** Confirms the local TCP/IP stack is intact and hasn't been corrupted. If this fails, the local machine's fundamental network security is compromised.

```
C:\Users\HP>ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```